UNIVERSITY OF THE PUNJAB

Roll No. .....

Course Code: CC-311-L **Time Allowed: 2 Hours**
Course: Operating Systems (Lab) **Max. Marks: 100**

Question # 1: **[25]**

Write a **Linux shell script** to demonstrate file and process management. Perform the following tasks:

1. Display the contents of the current directory using the ls command.
2. Create a directory named MyDirectory.
3. Inside this directory, create three files: **file1.txt, file2.txt, and file3.txt**.
4. Display the details of all files using **ls -l**.
5. Create a **hard link** and a **symbolic link** for file2.txt.

Question # 2: **[25]**

Write a C program demonstrating process synchronization where a parent process prints numbers from 6–10 and the child process prints 1–5 using **fork()** and **wait().**

Question #3: **[25]**

Write a multithreaded C program using **POSIX** threads.
Create two threads that print numbers from 1 to 5. Synchronize the threads using pthread_join().

Viva-Voce **[25]**